

# Cristian Lăcătuș

Software & automation engineer

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## SUMMARY

Final-year Automation & Applied Informatics student, software engineer intern at DeviDevs since July 2026. Most of what I build has no interface: Python jobs that run on a schedule, hold a file lock, retry a slow API and recover without a human. I like the front-end half too, so I am after a junior full-stack role.

## EXPERIENCE

### Software Engineer Intern — Automation & Integrations

Jul 2026 — present

DeviDevs Technologies · Bucharest (hybrid)

- **Human-in-the-loop agent bridge.** A Python daemon that lets me supervise a long coding-agent run from my phone or watch: verbose output summarised into one notification line and a question answerable in a single word, with a FIFO execution queue behind it. In daily use.
- **Single-writer concurrency.** An interactive session and a headless job write to the same Markdown vault, so writes are serialised with kernel file locks (`fcntl.flock`) and explicit signal handlers instead of hoping the two never overlap.
- **Scheduled automation and integrations.** Python ETL over Google Workspace APIs (Drive, Sheets, Gmail) running unattended under macOS `launchd`, with conditional HTTP caching and SQLite state, plus Playwright controllers driving Chrome over the DevTools Protocol where there is no API.
- **Guardrails for AI coding agents.** Lifecycle hooks that block an edit made after reading only a fraction of a file, plus a checker that verifies every documented claim against what is actually on disk — currently 537 file references and 29 capabilities, tested by running them.

### Academic practice — game systems and enemy AI

May — Jul 2026

DeviDevs Technologies / UTCB · team project

- Built the enemy AI and cover system for a 2D tank game in Unity and C#, in a team working on a normal branch, pull request and review workflow — roughly 87 hours logged in Jira across 10+ epics. Continued at the company as an intern afterwards.

### Operations & customer support

Jun — Oct 2022

Tucano Coffee · Bucharest — high-volume order flow, stock reconciliation and direct customer contact.

## SELECTED PROJECTS

### BetterTanks — enemy AI for a 2D tank game

Unity 2D · C#

- A five-state machine — patrol, chase, shoot, defensive, flee — where transitions are driven by health thresholds rather than distance alone: below 50% it looks for cover, below 25% it runs.
- No hand-placed cover markers: for every destructible wall still standing it derives four candidate positions from `Collider2D.bounds` and scores them, weighting proximity to itself ten times higher than distance from the player. The time went into behaviour, not structure — stopping it oscillating between two hiding spots, and handling the wall being destroyed while it is behind it. [Write-up](#) · [code on GitHub](#)

### Answering a coding agent from my watch

Python 3 · Unix daemons

- Telegram as transport, chosen for its native watchOS app; the hard part was compressing agent output into something readable on a wrist and a decision answerable in one word. Encryption between the three endpoints is not implemented yet, and the write-up says so. [Write-up](#)

### Portfolio site

HTML · CSS · Python

- Hand-written, no framework. Projects render from a single JSON source through a build script that refuses to publish when a page or a media file it claims does not exist on disk. [crs011.github.io](#)

## SKILLS

|         |                                                                                                                                         |
|---------|-----------------------------------------------------------------------------------------------------------------------------------------|
| CODE    | Python · C# (.NET) · JavaScript, HTML, CSS · Bash / Zsh · SQL · C / C++                                                                 |
| SYSTEMS | Unix daemons and <code>launchd</code> · POSIX file locking · Git branching, pull requests, review · REST APIs and OAuth · Docker · Jira |
| TOOLS   | Unity 2D · Playwright / Chrome DevTools Protocol · Google Cloud APIs · SQLite · Obsidian · AI coding agents                             |
| SPOKEN  | Romanian (native) · English (professional working proficiency, B2–C1)                                                                   |

## EDUCATION

### BSc, Automation & Applied Informatics (AIA)

2023 — 2027

Faculty of Hydrotechnical Engineering and Water Resources Management, Technical University of Civil Engineering of Bucharest (UTCB)

- Control systems, differential equations, signal analysis, instrumentation, industrial automation, software engineering. Final year.
- EU-CONEXUS micro-credentials, 2024: Neuroscience & AI (Catholic University of Valencia) · Film Literacy (University of Zadar).